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CAREER HIGHLIGHTS

Software Engineer specializing in software systems, backend development, and application solutions (C++, Python, and cloud systems). Contributed to developing cloud-native software components and workflows across AWS, Azure, and Kubernetes. Applied software development, testing, CI/CD pipelines, and best coding practices.

Focused on designing, building, and delivering reliable software systems from PoC to production.

PROFESSIONAL EXPERIENCE

10/2025 – 04/2026
Bonn, Germany

Deutsche Telekom 🌐, *Software Engineer | AI Platforms & MLOps*

- Built a microservices-based LLM benchmarking platform on vLLM using GuideLLM (latency, throughput), BFCL (tool calling), and DeepEval (LLM-as-judge), via Streamlit.
- Consolidated experiment tracking and SLO observability in ClearML, enabling teams to make data-driven model selection and capacity planning across LLM deployments.
- Automated large-scale benchmarking through GitLab CI/CD and Kubernetes, executing matrix jobs on Hugging Face datasets and publishing metrics to Grafana.
- Hardened enterprise MLflow on Kubernetes using Keycloak OIDC, OAuth sidecars, TLS and RBAC, enforcing platform governance and compliance on the model registry.

05/2025 – 09/2025
Wolfsburg,
Germany

Volkswagen Group (CARIAD) 🌐, *Master Thesis | Integrating LLMs In Automotive Systems*

- Led the development of AI-driven White Box fuzzing for automotive compute platforms, automating test generation and improving code flaw detection by 13%.
- Benchmarked 16 LLMs and applied LoRA/QLoRA fine-tuning to optimize model performance, reducing token consumption by 55% and test generation time by 33%.
- Designed a secure Azure Private Link architecture for LLM workflows, enabling CI/CD runners without internet access to securely access inference pipelines in production.
- Integrated autonomous AI test generation workflow in CI/CD/CT pipelines via GitHub Actions, boosting line and branch coverage by 7% to improve security validation.

10/2024 – 04/2025
Munich, Germany

BMW Group 🌐, *Data Scientist Intern | RAG & LLM Systems*

- Deployed a CRNN surface detection model on AWS SageMaker & S3 with semantic segmentation, tuning AI parameters and serving results via API to cut false positives.
- Architected a Django Web application for parsing geometric STL meshes and file ingestion, utilizing document chunking pipelines to achieve 8% faster processing.
- Engineered an LLM auto-suggestion model utilizing RAG and vector search to identify part issues, via document chunking and data indexing to elevate precision by 12%.
- Optimized PySpark pipelines on AWS EMR for surface defect data, integrating new defect records and restructuring data schemas for QuickSight reporting dashboards.

10/2021 – 09/2022
Chennai, India

Crystal Consultancy Services 🌐, *Data Analyst*

- Connected SAP R3 to Power BI to design automated regional sales dashboards, replacing manual Excel reporting to standardize weekly KPIs, saving 4 hrs/week.
- Refactored PySpark transformations in Azure Databricks to eliminate redundant joins and bottlenecks in SAP ETL workflows, improving pipeline throughput by 25%.
- Explored data modeling in dbt on SAP sales data, defining modular SQL models to standardize regional KPI calculations and ensure consistent KPI aggregations.

02/2021 – 07/2021 Texas, USA (Remote)	Altascio Technologies  , <i>Data Engineering Intern</i> Consolidated fragmented inventory data using SQL joins and aggregations, improving data consistency and saving 20 hrs/week in manual operational data updates.
06/2020 – 12/2020 London, United Kingdom (Remote)	Jman Group , <i>Python Developer Intern</i> Assisted in developing predictive models on real-time sensor data to identify clusters, detect outliers, and uncover trends, achieving 87% model accuracy in field tests.

TECHNICAL SKILLS

Programming: Python (PyTorch, TensorFlow, Scikit-learn, FastAPI) Go C++ TypeScript Computer Vision
Data Engineering & Analytics: SQL PostgreSQL, Snowflake dbt, PySpark ETL/ELT, Data Warehousing
GenAI & LLM Systems: RAG, AI Agents, LLM Evals LangChain, LangGraph, vLLM, Ollama, DeepEval LoRA, MCP, Hugging Face Vector DBs, Embeddings, Semantic Search, OpenSearch AI Governance, EU AI Act
Cloud & MLOps: AWS (EC2, EKS, ECR, Lambda, Bedrock, SageMaker) Docker, Kubernetes (Helm), Redis MLflow, Airflow, ClearML OpenTelemetry, Grafana GitHub Actions, GitLab CI/CD Git, DVC Linux, Bash

EDUCATION

10/2022 – 03/2026	M.Sc Data Science , <i>Friedrich Alexander University Erlangen-Nuremberg</i>  Subjects: NLP, Deep & Reinforcement Learning, Pattern Recognition, Statistics, Finance
04/2018 – 01/2022	B.E. in Computer Science Engineering , <i>Anna University, India</i> 

PROJECTS

Inference Control Plane  , <i>LLM Infrastructure & Routing System</i> Developed a production-grade FastAPI gateway for LLM traffic control with priority-based routing, fallback routing, token optimization, Redis semantic caching, per-key rate limiting, and PostgreSQL audit logging.
GuardrailX  , <i>Enterprise AI Governance Platform</i> Enforced runtime LLM governance across enterprise systems through prompt injection detection, jailbreak detection, PII redaction, content safety filtering, hallucination controls, and risk-based policy enforcement.
AI Infrastructure Hypervisor  , <i>GPU-Aware Virtualization Platform</i> Constructed a GPU-aware AI runtime using KVM/QEMU to provision isolated vLLM workloads with automated orchestration, Prometheus/Grafana observability, and runtime scheduling across inference environments.
IntentGraph  , <i>Agentic AI Planning & Orchestration System</i> Compiled natural-language goals into policy-gated DAG workflows across a multi-tenant agentic OS with schema validation, human approval gates, audit tracing, scoped runtime memory, and real-time replanning.
Scientific AI Mesh Platform  , <i>Graph Neural Network System</i> Implemented a heterogeneous GNN framework combining GAT, GraphSAGE, and GCN with edge-aware attention to predict structural wear at mesh nodes from raw geometric and material properties (RMSE < 0.001).

CERTIFICATIONS

Predictive Data Analytics for Business Intelligence Nanodegree  Excel to MySQL: Regression  Statistics 
DL AI TensorFlow Developer  Google Cloud Platform IT Automation  Data Warehousing, Big Data on GCP 

LANGUAGES

English: C1 (Business fluent) German: A2 erworben, B1 laufend  French: A1  Tamil: Native
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